GGG-P-676a
January 5, 1965

SUPERSEDING
Fed. Spec. GGG-P-676
May 17, 1951

FEDERAL SPECIFICATION

PROTRACTORS, MECHANIC'S, BEVEL AND PLAIN

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. The protractors covered by this specification are designed for precision measuring, checking and laying out of angular work, and scribing layout lines.

1.1.1 *Federal specification coverage.* This specification does not include all types, classes, and styles of protractors that are commercially available, but only those most commonly used by the Federal Government.

1.2 Classification.

1.2.1 *Types, classes, and styles.* Protractors covered by this specification shall be furnished in the following types, classes, and styles, as specified (see 6.2):

Type I—Protractor, universal bevel, vernier scale.
- Class 1—Revolving dial.
  - Style A—Without acute-angle attachment.
  - Style B—With acute-angle attachment.
- Class 2—Stationary dial.
  - Style A—Without acute-angle attachment.
  - Style B—With acute-angle attachment.

Type II—Protractor, bevel, non-vernier.

Type III—Protractor, plain.
- Style A—Half-round head.
- Style B—Rectangular head.

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 Specifications and standards. The following specifications and standards, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification.

Federal standards:

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.
Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

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Military Specification:

MIL-H-15424—Hand Tools; Packaging of.

Military Standards:

MIL-STD-105—Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129—Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 **Other publication.** The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

National Bureau Of Standards Handbook:
H28—Screw Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D. C. 20402.)

3. REQUIREMENTS

3.1 **Illustrations.** The illustrations shown herein are descriptive and not restrictive, and are not intended to preclude the purchase of protractors which are otherwise in accordance with the requirements of this specification.

3.1.1 *Materials.* The materials used and heat treatment employed shall be such as to produce protractors capable of long life under ordinary service conditions and of withstanding specified tests.

3.2 **Design.** Protractors shall be suitable for the use intended and, for type I, shall be composed essentially of a vernier scale, frame, graduated dial, thumb nuts, and adjustable blades. Type II protractors shall be essentially the same except that they are not provided with a vernier scale. Type III shall have a half-round or rectangular head with graduations and a non-adjustable blade. All screw threads shall conform with the current issue of Screw-Thread Standards for Federal Services, NBS Handbook H28, and shall have a class 2 fit or better.

3.3 **Finish.** All exterior surfaces of instruments covered by this specification shall have a finish such that the surface roughness does not exceed 32 micro-inches r.m.s.

3.4 **Threads.** All screw threads shall conform to H28 and shall be a class 2 fit or better.

3.5 **Identification marking.** Protractors shall be marked in a permanent and legible manner with the manufacturer's name or trademark of such known character that the source of manufacture may be readily determined.

3.6 **Instruction leaflet.** Each protractor, except type III, shall be accompanied with a clear and concise operational instruction leaflet.

3.7 **Type I, class 1, revolving dial, style A.** Type I, class 1, style A, revolving dial universal bevel protractors shall consist essentially of a vernier scale, frame, revolving dial, thumb nuts, and adjustable blades. Type I, class 1, style A protractor shall be similar to figure 1.

3.7.1 *Vernier scale.* The vernier scale shall be made to fit neatly on the beveled edge of the frame and shall be held rigidly in position; the graduations shall extend to the outer surface edge of the scale. It shall be made and fitted so as to minimize parallactic error in reading between the vernier scale and the main scale. There shall be a clearance of not more than 0.002 inch between the bottom surface of the vernier scale and the surface of the main scale directly beneath the vernier scales' outer lower edge. The vernier scale shall be graduated in 5 minute (5') increments from 0 minutes up to 60 minutes

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both right and left from the zero index. The graduations shall be cut clearly and straight to a width of not less than 0.003 inch nor more than 0.006 inch and to a depth of not less than 0.001 inch nor more than 0.003 inch. The 0 minute and each succeeding 15-minute graduation shall be marked with the corresponding number and shall be longer than the intermediate graduations. The intermediate graduations shall be readable to 5 minute increments. The vernier scale shall be accurate within ± 1 minute between any two graduations.

3.7.2 *Frame.* The frame shall be of one-piece construction and so made that the protractor will lie flat upon the work. It shall consist essentially of a circular member upon which is mounted the vernier scale, and a base whose top and bottom surfaces are at right angles to the diameter of the circular section passing through zero graduation on the vernier scale. At zero setting of the protractor, measuring edges of blades and base shall be parallel within ± 2-1/2 minutes. Measuring edges shall be flat within 0.0005 inch over the length of the edges. The frame shall have a hardness value of not less than 92 on the Rockwell B scale.

3.7.3 *Dial.* The dial shall be not less than 2-1/2 inches nor more than 3 inches in diameter and shall have on the outer periphery of the front surface a scale which is accurately graduated in degrees. It shall be rotatable in a complete circle upon a center stud. The scale shall be divided into four quadrants each having 90 graduations. The zero and each 10° graduation in either direction shall be marked with the corresponding numeral. The 5° graduations shall be longer than the intermediate graduations. A tolerance of ± 2-1/2 minutes noncumulative between any two graduations shall be allowed. Width and depth of graduation lines shall be as specified for vernier scale (see 3.7.1).

3.7.4 *Thumb nuts.* Thumb nuts or other devices shall be furnished for locking the rotating dial to the frame and locking the blade at any point.

3.7.5 *Setting.* A positive means shall be furnished for securing fine adjustments and smooth movement of the dial in either direction.

3.7.6 *Adjustable blades.* Adjustable blades shall be capable of being moved smoothly back and forth through their entire length and shall be interchangeable. They shall have a longitudinal groove parallel with the edges to facilitate their being clamped to the revolving dial and shall have a hardness value of not less than 92 on the Rockwell B scale. The ends of the blades shall be machined to angles with one blade edge of 45° on one end and 60° on the other end. Blades of 12 inches or 6-1/2 inches ± 1/2 inch shall be furnished as specified (see 6.2). Blades shall be flat and parallel within 0.0005 inch.

3.8 Type I, class 1, revolving dial, style B. The style B protractor shall be essentially the same as the style A, and be similar to figure 1, with the exception that an angle attachment shall be furnished.

3.8.1 *Acute-angle attachment.* The acute-angle attachment shall be made so as to fit readily on the base. It shall be capable of being clamped at any desired point within the limit provided in the base. When clamped, its longitudinal axis shall be perpendicular within ± 2' to the base and in the same plane as the blade.

3.9 Type I, class 2, stationary dial, style A. Type I, class 2, style A, stationary dial universal bevel protractors shall consist essentially of a vernier scale, frame, revolving plate, thumb nut, and adjustable blades. Type I, class 2, style A protractors shall be similar to figure 2.

3.9.1 *Vernier scale.* The vernier scale shall be made to fit neatly on the revolving plate and shall conform with requirements, other than location, given in 3.7.1.

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3.9.2 *Frame.* The frame shall be of one-piece construction and so made that the protractor will lie flat upon the work. It shall consist essentially of a circular member upon which appear the dial graduations, a base at right angles to the diameter of the dial passing through the zero graduation, and a fixed arm extending at right angles to the bottom working edge of the base. At zero setting of the protractor, measuring edges of blade and base shall be parallel within ± 2 minutes. The frame shall have a hardness value of not less than 96 on the Rockwell B scale.

3.9.3 *Dial.* The dial shall be not less than 2-1/2 inches nor more than 3 inches in diameter and shall have on the outer periphery of the front surface a scale which is accurately graduated in degrees. The scale shall be divided into four quadrants each having 90 graduations. The zero and each 10° graduation in either direction shall be marked with the corresponding numerals. The 5° graduations shall be longer than the intermediate graduations. Angular tolerance shall conform to that given in 3.7.3, and width and depth of graduations shall be as specified in 3.7.1.

3.9.4 *Base.* The base shall be substantially rectangular in section and shall be not less than 3-3/4 inches nor more than 4-1/2 inches in length. The bottom edge shall be flat and straight within 0.0005 inch. The front and back faces shall be straight, flat, and parallel within 0.0005 inch. A portion of the base shall extend beyond the fixed arm and the end shall be straight, flat, and at right angles to all other base surfaces.

3.9.5 *Fixed arm.* The fixed arm shall have edges at right angles to the bottom working edge and shall be not less than 2-1/2 inches nor more than 2-3/4 inches in height and shall be set off far enough from the circular section so as not to interfere with the reading of the dial or with blade movement and shall be of suitable shape for acute-angle attachment.

3.9.6 *Revolving plate.* The revolving plate shall be rotatable in a complete circle and shall be provided with means for holding and locking the blade. Provision shall also be made for the mounting of the vernier scale in an appropriate position upon the revolving plate.

3.9.7 *Thumb nut.* At least one thumb nut measuring not more than 1-3/16 inches in diameter shall be furnished for easy and rigid clamping of the revolving plate to the dial.

3.9.8 *Setting.* A positive means shall be furnished for securing fine adjustments and smooth movement of the dial in either direction.

3.9.9 *Adjustable blades.* Adjustable blades shall comply with the requirements of 3.7.6.

3.10 **Type I, class 2, stationary dial, style B.** The style B protractor shall be essentially the same as style A, and be similar to figure 1a, except that an acute-angle attachment shall be furnished.

3.10.1 *Acute angle attachment.* The acute angle attachment shall fit readily on the base. It shall be capable of being clamped at any point within the limit provided in the base. When clamped, its longitudinal axis shall be parallel within ± 2 minutes to the base and in the same plane as the base.

3.11 **Type II, bevel, non-vernier.** Type II, non-vernier, revolving-dial bevel protractors shall consist essentially of a frame, revolving dial, thumb nut, and adjustable blades. Type II protractors shall be similar to figure 3.

3.11.1 *Frame.* The frame shall be of one-piece construction and so made that the protractor will lie flat upon the work. It shall consist essentially of a circular member with three zero graduations spaced at 90° around the dial and centered at top and a base with top and bottom surfaces at right angles to the diameter of the circular member passing through the top zero gradua-

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tion. At zero setting of the protractor, measuring edges of blade and base shall be at right angles within ± 5 minutes. The frame shall have a hardness value of not less than 96 on the Rockwell B scale.

3.11.2 *Dial.* The dial shall be not less than 1-3/4 inches nor more than 2-1/2 inches in diameter and shall be graduated on the outer periphery of the front surface. It shall be rotatable in a complete circle upon a center stud and be provided with means of holding and locking the dial and blade. The scale shall be graduated in degrees over an arc of 180°, reading from 0° to 90° from each extremity of the arc. The zero and each 10° graduation in either direction shall be marked with the corresponding numeral. The 5° graduations shall be longer than the intermediate graduations. Angular tolerance shall conform to that as shown in 3.7.3, and width and depth of graduations shall be as specified in 3.7.1.

3.11.3 *Base.* The base shall be substantially rectangular in section and shall be not less than 3 inches nor more than 4-1/2 inches in length. The working edges at top and bottom shall be straight, flat, and parallel within 0.0005 inch. The end of the base shall be straight, flat, and at right angle to all other base surfaces.

3.11.4 *Thumb nuts.* Thumb nuts or other devices shall be furnished for locking the rotating dial to the frame and locking the blade at any point.

3.11.5 *Adjustable blades.* Adjustable blades shall comply with the requirements of 3.7.6.

3.12 Type III, plain, style A. Half-round head protractors shall consist essentially of a half-round head with double graduations reading both ways, blade and lock-joint and be similar to figure 4.

3.12.1 *Half-round head.* The half-round head shall be not less than 3-1/4 inches nor more than 3-3/8 inches in length and not less than 1-7/8 inches nor more than 2 inches in height. There shall be one finished working edge on the head. This edge shall be parallel to an imaginary line through opposite 0° and 180° graduations It shall have two scales accurately graduated in degrees, reading both ways to show the complementary degrees from 0° to 180° on each scale, and shall be accurate within plus or minus 5 minutes between any two graduations. The surface of the working edge shall be flat within 0.0005 inch.

3.12.2 *Blade.* The indicating arm of the blade shall have a line graduation for accurate setting and reading of the protractor. The opposite end of the blade shall be not less than 6 inches nor more than 6-1/4 inches in length from the center of the stud hole. The blade shall be approximately 5/8 inch in width and 1/16 inch in thickness with working edges accurately straight within 0.0005 inch and parallel within 0.0005 inch. The blade shall be mounted on the head in such manner as to provide free and accurate positioning.

3.12.3 *Thumb nut.* A knurled thumb nut or other locking device shall be furnished for giving adjustable tension and locking the blade at any point.

3.13 Type III, plain, style B. The rectangular head protractor shall be essentially the same as style A, with the exception that the head shall be rectangular in shape and shall be not less than 3-1/4 inches nor more than 3-3/8 inches in length and not less than 1-7/8 inches nor more than 2 inches in width. The head shall have four finished working edges, the surfaces of which shall be flat, parallel, and square, within 0.0005 inch. Style B protractors shall be similar to figure 5.

3.13.1 *Blade.* The blade shall conform to requirements of 3.12.2.

3.13.2 *Thumb nut.* The thumb nut shall conform to requirements of 3.12.3.

3.14 Workmanship. Workmanship of the protractors shall be of the quality prevail-

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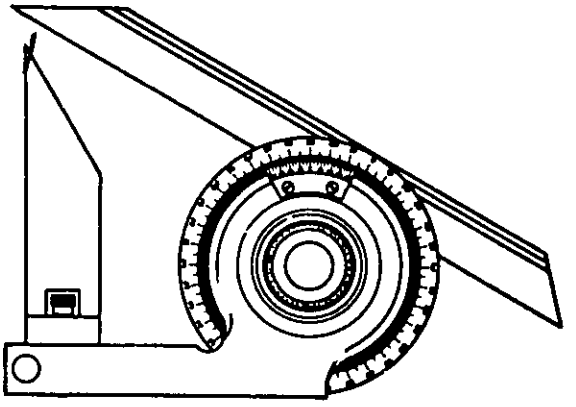


FIGURE 1.—Type I, class 1, protractor, universal bevel, revolving dial.

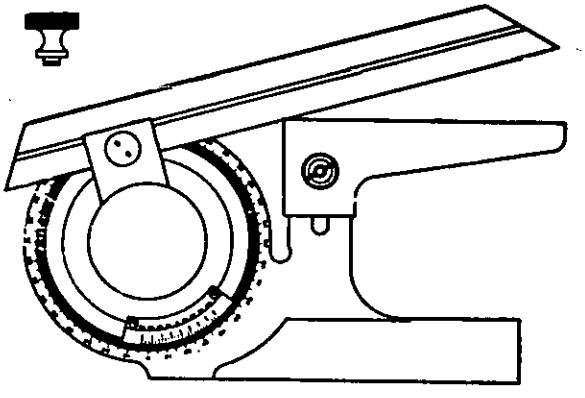


FIGURE 2.—Type I, class 2, protractor, universal bevel, stationary dial

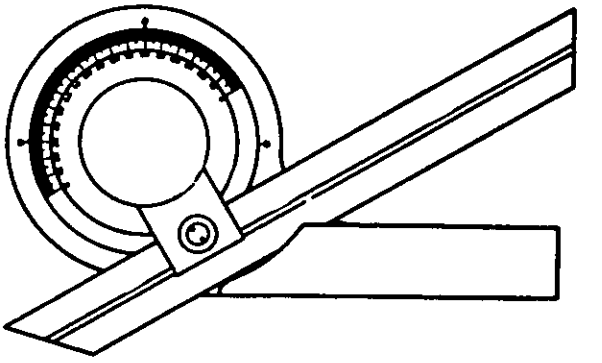


FIGURE 3.—Type II protractor, bevel, non-vernier.

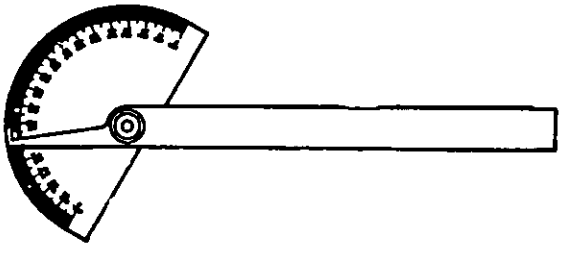


FIGURE 4.—Type III, style A, protractor, plain, half round head.

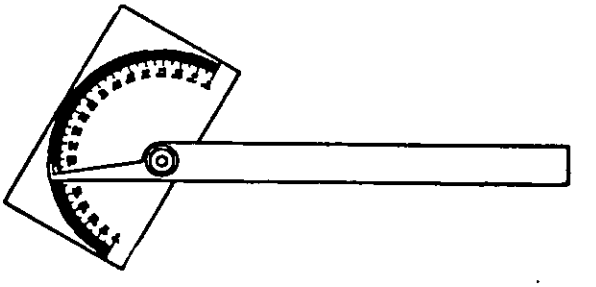


FIGURE 5.—Type III, style B, protractor, plain, rectangular head.

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ing among manufacturers normally producing protractors of the types specified herein.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Sampling for quality conformance inspection.

4.2.1 *Lot.* All protractors of the same type, class and size presented at one time shall be considered a lot for purposes of quality conformance inspection.

4.2.2 *Sampling.* A random sample of protractors shall be selected from each lot in accordance with MIL-STD-105 at inspection level II.

4.2.3 *Quality conformance inspection.* Quality conformance inspection shall be performed on each protractor selected in accordance with 4.2.2. Quality conformance inspection shall consist of the following:

a. Examination (4.3).
b. Tests (4.4).
c. Inspection of preparation for delivery (4.5).

4.3 Examination. Each of the sample protractors selected in accordance with 4.2.2 shall be examined to verify compliance with this specification. The acceptable quality level (AQL) shall be 2.5 percent defective. Any protractors in the sample containing one or more defects shall be rejected, and if the number of defective protractors in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected.

4.4 Test procedures.

4.4.1 *Accuracy tests.* Each sample shall be tested for compliance with the accuracy requirements applicable to each individual type and class. The AQL for tests shall be 1.5 percent defective. Nonconforming protractors shall not be offered for delivery and if the number of nonconforming protractors exceeds the acceptance number for the samples tested, the lot represented shall be rejected.

4.4.2 *Test temperature.* Tests shall be conducted at 68° ± 2°F., if practicable. Calibrations at other temperatures shall be converted to the standard temperature of 68°F. All test equipment and protractors tested shall be at the same constant temperature.

4.4.3 *Test apparatus and instruments.* Tests for accuracy shall be made with suitable equipment of proven accuracy.

4.4.3.1 Flatness shall be determined with a test indicator used in conjunction with a surface gage and surface plate.

4.4.3.2 Squareness shall be determined by means of a precision square and gage blocks.

4.4.3.3 Angular intervals shall be determined by means of either a circular measuring machine or a sine bar or angular blocks to ascertain the total angular error.

4.4.4 *Accuracy of test determination.* Where the permissible tolerance specified herein for any protractor or its component members is 0.0005 inch or larger, the error in any measurement for accuracy shall not exceed 20 percent of the permissible tolerance.

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4.5 **Inspection of preparation for delivery.** Sample protractors and packages shall be selected and inspected to verify conformance to the requirements of section 5.

5. PREPARATION FOR DELIVERY

(For civil agencies, Fed. Std. No. 102 should be referred to for definitions and applications of the various levels of packaging protection for supplies and equipment.)

5.1 **Preservation, packaging, and packing.** Preservation, packaging, and packing shall be in accordance with MIL-H-15424. Levels of preservation and packaging and level of packing shall be as specified (see 6.2).

5.2 **Marking.**

5.2.1 *Military agencies.* In addition to any special marking required (see 6.2), marking shall be in accordance with MIL-STD-129.

5.2.2 *Civil agencies.* In addition to any special marking required (see 6.2), marking for shipment shall be in accordance with Fed. Std. No. 123.

6. NOTES

6.1 **Intended use.** Types I and II protractors are intended to be used by craftsmen for scribing, checking, and measuring angular work where extreme accuracy is necessary. Type III protractors are designed for checking angular surfaces not requiring extreme accuracy.

6.2 Ordering data. Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(*a*) Title, number, and date of this specification.
(*b*) Type, class, and style of protractor (see 1.2.1).
(*c*) Length of blade required for types I and II (see 3.7.6).
(*d*) Applicable levels of preservation, packaging, and packing (see 5.1).
(*e*) Special marking, if required (see 5.2).

6.3 **Transportation description.** Transportation descriptions and minimum weights applicable to this commodity are:

Rail:

Tools, not otherwise indexed by name.
Carload minimum weight 30,000 pounds.

Motor:

Tools, not otherwise indexed.
Truckload minimum weight 30,000 pounds, subject to Rule 115, National Motor Freight Classification.

MILITARY CUSTODIANS:

Army—GL

Navy—SH

Air Force—67

Review activities:

Army—GL

Navy—SH

Air Force—67

User activities:

Navy—MC, YD

Preparing activity:

Navy—SH

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